# YUNHAI XIANG

Mathematician · Data Scientist · Software Engineer

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## PROFESSIONAL SUMMARY

I am a pure mathematician by training, with extensive experience in formal and quantitative reasoning, communicating technical concepts through teaching and presentations, and software engineering.

## EDUCATION

**Western University (University of Western Ontario)** *2024 — Now*
*Ph.D. (in progress) and M.Sc. in Mathematics* *London, ON, Canada*
Advisor: Chris Hall
Specialization: Number Theory and Arithmetic Geometry

**University of Waterloo** *2018 — 2023*
*B.Math. in Pure Mathematics, Honors, with Distinction* *Waterloo, ON, Canada*

## INDUSTRY EXPERIENCE

**ByteDance** 2019
*Co-op Internship* *Beijing, China*

- Supported data-informed product development for an education technology product
- Collected and reviewed educational data to support product analysis and application improvement

## SKILLS

*Colored by level of mastery: Expert, Advanced, Proficient, Familiar, Beginner*

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| **Languages** | Mandarin, English, French, Japanese |
| **Programming** | Python, SQL, Lean, C/C++, JavaScript/TypeScript, C#, Rust, Go |
| **DS/ML** | NumPy, SciPy, Pandas, PyTorch, XGBoost, Sklearn |
| **Graphics** | Cinema 4D, Blender, Unity, Godot |
| **Hobbies** | Magic Tricks, Game Development, Jazz Music, Philosophy, Cooking |

## PUBLICATIONS & PREPRINTS

*QEDBench: Quantifying the Alignment Gap in Automated Evaluation of University-Level Mathematical Proofs*
Joint with Santiago Gonzalez, Alireza Amiri Bavandpour, Peter Ye, Edward Zhang, et al.
INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) 2026

## AWARDS

**Mathematics Undergraduate Research Award** 2022
*University of Waterloo*

- Highly competitive undergraduate research award of value $10000 at University of Waterloo. Participated in research project related to Weyl groups, supervized by Prof. Matthew Satriano.

## CONFERENCES & SEMINARS

**Conferences Attendance & Presentations**
*Attended academic conferences and delivered research or expository talks*

- ∞-Type Café Summer School 2026, Online
- FrontierMath Open Problems Workshop, Summer 2026, UToronto
- Homotopy Theory, K-theory, and Topological Data Analysis Conference, Summer 2026, Western
- Using AI to Find Examples Workshop, under thematic program Universal Statistics in Number Theory, Winter 2026, Centre de Recherches Mathématiques, Université de Montréal
- University of Western Ontario Research in Computer Science (UWORCS), Winter 2026, Western
- *Rigid Analytic Geometry*, Ontario Graduate Mathematics Conference, Summer 2024, UWaterloo
- *Cohomology of Grassmannians*, Canadian Undergraduate Mathematics Conference, Summer 2022, Université Laval

**Seminar Attendance & Presentations**
*Attended academic seminars and delivered research or expository talks*

- *Hyperelliptic Jacobians and Hasse–Witt Matrix*, Algebraic Geometry Student Seminar, Western
- *Trace of Frobenius*, Winter 2026, Graduate Student Seminar, Western
- *Brauer Groups of Fields Part II*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Brauer Groups of Fields Part I*, Winter 2026, Algebraic Geometry Student Seminar, Western
- *Weyl Group and Root Datum*, Summer 2025, Algebraic Groups Seminar, Western
- *Hecke Operators*, Summer 2025, Algebraic Groups Seminar, Western
- *Parabolic and Borel Subgroups*, Summer 2025, Algebraic Groups Seminar, Western
- *Lie Algebras of Algebraic Groups*, Summer 2025, Algebraic Groups Seminar, Western
- *Admissible Topology*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Non-Archimedean Fields*, Summer 2024, Rigid Analytic Geometry Seminar, Western
- *Cones and Affine Toric Varieties*, Summer 2024, Toric Varieties Learning Seminar, Western
- *An Easy Tour of Galois Cohomology*, Winter 2024, Graduate Student Seminar, Western
- *Introduction to Derived Categories*, Fall 2022, Algebraic Geometry Working Seminar, UWaterloo

## TEACHING & MENTORING

**Teaching Assistantship**
*Proctored and graded exams, graded homeworks, tutored at Help Centre, and taught tutorial sessions*

- MATH 3150, *Elementary Number Theory I*, Fall 2024
- MATH 2155, *Mathematical Structures*, Fall 2025, Fall 2026
- CALCULUS 1500, *Calculus I for the Mathematical Sciences*, Fall 2026
- CALCULUS 1501, *Calculus II for Mathematical and Physical Sciences*, Winter 2025, Winter 2026, Summer 2026
- MATH 1229, *Methods of Matrix Algebra*, Summer 2024, Summer 2025
- MATH 1228, *Methods of Finite Mathematics*, Summer 2024, Summer 2025
- CALCULUS 1000, *Calculus I*, Winter 2024, Summer 2024, Winter 2025, Fall 2025, Winter 2026

**Mathematical Outreach Lectures**
*Gave expository lectures to high school students or undergraduate students on advanced mathematical topics*

- *Aperiodic Tilings*, Summer 2026, Western Summer Math Camp (Grade 11/12)
- *Surreal Numbers and the Game of Go*, Summer 2025, Western Summer Math Camp (Grade 10)
- *Mathematical Magic Tricks: Gilbreath's Principle*, Summer 2024, Western Summer Math Camp (Grade 11)
- *Why is There no Quintic Formula*, Summer 2024, Western Summer Math Camp (Grade 10)
- *Category Theory Demystified*, Winter 2023, Short Attention Span Math Seminar (Undergraduate)

**Directed Reading Program Mentorship**
*Mentored undergraduate students on reading projects, culminating in a presentation*

- Jude McNorgan Lubos, *Introduction to Quadratic Reciprocity*, Winter 2026
- Arvin Babakri, *Hensel's Lemma in p-adic Analysis*, Fall 2026

## SELECTED SOFTWARE PROJECTS & OPEN SOURCE CONTRIBUTIONS

**Games and Graphics Softwares**

- Coming Soon.

**Mathematics and Scientific Computing**

- Coming Soon.

## CERTIFICATIONS

**Language Proficiency**

- Coming Soon.

**Boot Camps & Programs**

- Coming Soon.

**Technical Certifications**

- Coming Soon.

## SERVICE AND VOLUNTEERING

**Seminar and Conference Organization**

- Graduate Student Research Seminar, Fall 2026, co-organized with Alex Zwart
- Algebraic Geometry Student Seminar, Summer 2026
- Algebraic Geometry Student Seminar, Winter 2026, co-organized with Alex Zwart
- Algebraic Geometry Student Seminar, Fall 2024, co-organized with Alex Zwart
- Toric Varieties Learning Seminar, Summer 2024, co-organized with Nikolas Rieke